

## Extra — Creating Dictionaries: Add & Update

**Topic:** Building & Changing a Dictionary · **Course:** Extra Worksheet · **Time:** about 35 minutes

You can start with an empty dictionary and fill it later, and you can change a value after you set it.

`player["score"] = 10` adds a new key, or updates one that already exists. On paper you will read, predict, fix, and explain. Then you will write and run your own code in the IDE.

Keep these words handy: **key**, **value**, **add a key**, **update**, **dict[key] = value**.

### 1 · Predict 🧠

Read each snippet. Write what you think it will print.

```
player = {"name": "Leo"}
player["score"] = 0
player["score"] = 50
print(player["score"])
```

**What prints? The score was set twice — which value wins?**

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```
pet = {}
pet["kind"] = "dog"
pet["age"] = 4
print(pet)
```

**What does the whole dictionary look like when it prints?**

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```
car = {"color": "red", "speed": 100}  
car["speed"] = car["speed"] + 20  
print(car["speed"])
```

**What prints? Work out the right side first, then the left.**

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## 2 · Spot the Bug 🐛

Each snippet is broken. Read what it should do, then rewrite it so it works. Explain why the original was wrong.

**Bug A** — This should add a `"level"` key to the player. Instead it rebuilds the whole dictionary and loses the name.

```
player = {"name": "Leo"}  
player = {"level": 1}  
print(player["name"])
```

**Hint:** to add a key, set `player["level"] = 1` — don't make a brand-new dictionary.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — This should raise the score by `100`. Right now it crashes, because it reads a key that was never created.

```
game = {"lives": 3}  
game["score"] = game["score"] + 100
```

**Hint:** you can't add to a `"score"` that doesn't exist yet. Give it a starting value first.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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### 3 · Explain the Code

```
profile = {}
profile["username"] = "coder_kim"
profile["wins"] = 0

profile["wins"] = profile["wins"] + 1
profile["wins"] = profile["wins"] + 1

print(f"{profile['username']} has {profile['wins']} wins")
```

**profile** starts empty. What is inside it after the first two lines?

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The two **wins** lines look almost the same. What do they do to **wins** each time?

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What is the final printout, and why is **wins** **2** and not **1**?

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## 4 · Code It

Open the IDE at **app.english-coding.co.uk** and write your own program.

Write code that: - makes an **empty dictionary** called `character` - adds a `"name"` key and an `"hp"` key with values you choose - then **updates** `"hp"` to a new value - prints the character's name and final HP

Run it, then write your code and what it printed.

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**Submit** 

Send this worksheet + a screenshot of your working code to teacher on KakaoTalk.